

UNIT I

1. What is the critical section in process synchronization?

- a) A segment of code where shared resources are accessed**
- b) A method of process communication**
- c) A section of a program that never accesses shared data**
- d) A technique to prevent deadlock**

Answer: a

2. Which of the following is true about Peterson's solution?

- a) It uses semaphores for process synchronization**
- b) It allows multiple processes to enter the critical section simultaneously**
- c) It works for only two processes**
- d) It cannot guarantee mutual exclusion**

Answer: c

3. What does synchronization hardware typically do?

- a) Ensures that only one process can enter the critical section at a time**
- b) Detects deadlocks automatically**
- c) Allocates memory to processes**
- d) Enables swapping between processes**

Answer: a

4. Which of the following is a property of semaphores?

- a) Semaphores can only be used for deadlock detection**

- b) Semaphores are a signaling mechanism
- c) Semaphores do not ensure mutual exclusion
- d) Semaphores are used to swap processes

Answer: b

5. In process synchronization, which problem involves readers and writers trying to access a shared database?

- a) Producer-consumer problem
- b) Dining philosophers problem
- c) Readers-writers problem
- d) Banker's algorithm

Answer: c

6. What is a system model in deadlock characterization?

- a) A method for deadlock prevention
- b) A representation of processes and resources as a graph
- c) A scheduling algorithm for processes
- d) A technique for synchronization

Answer: b

7. Which of the following is a method for deadlock prevention?

- a) Allow circular wait conditions
- b) Ensure at least one process can acquire all resources
- c) Allow processes to request resources only in a specific order
- d) Make processes wait indefinitely

Answer: c

8. What does deadlock avoidance involve?

- a) Detecting deadlocks after they occur**
- b) Allocating resources dynamically based on current availability**
- c) Swapping processes to avoid deadlocks**
- d) Killing processes to avoid deadlocks**

Answer: b

9. In memory management, what does paging do?

- a) Allocates entire memory to a single process**
- b) Divides the process memory into equal-sized pages**
- c) Prevents deadlocks from occurring**
- d) Ensures processes access memory in sequential order**

Answer: b

10. What is the purpose of the page table in a paging system?

- a) To track free memory**
- b) To map virtual addresses to physical addresses**
- c) To detect deadlocks**
- d) To manage process communication**

Answer: b

11. What is the main purpose of process synchronization?

- a) To prevent processes from entering the critical section
- b) To avoid deadlocks
- c) To ensure that multiple processes access shared resources correctly
- d) To enable process communication

Answer: c

12. In Peterson's solution, which variables are used to ensure synchronization?

- a) lock and signal
- b) turn and flag
- c) wait and signal
- d) semaphore and mutex

Answer: b

13. Which of the following is a classical problem of synchronization?

- a) Page replacement problem
- b) Dining philosophers problem
- c) Paging and segmentation
- d) Priority inversion problem

Answer: b

14. What is the main disadvantage of Peterson's solution?

- a) It cannot guarantee mutual exclusion
- b) It only works for multiple processes
- c) It is not suitable for modern multiprocessor systems
- d) It does not prevent deadlocks

Answer: c

15. What is a mutex lock?

- a) A software-based synchronization technique**
- b) A hardware-based memory management technique**
- c) A type of semaphore**
- d) A scheduling algorithm**

Answer: a

16. How do semaphores differ from mutexes?

- a) Semaphores are used for single-threaded applications**
- b) Mutexes can have a value greater than 1**
- c) Semaphores can be used to signal multiple processes**
- d) Mutexes are software-based while semaphores are hardware-based**

Answer: c

17. What is the key idea behind the Banker's algorithm for deadlock avoidance?

- a) Processes must wait for resources in a circular manner**
- b) Resources are allocated dynamically to prevent deadlocks**
- c) All resources must be pre-allocated to processes**
- d) It ensures the system always remains in a safe state**

Answer: d

18. Which condition must hold for deadlock to occur?

- a) Preemption
- b) Circular wait
- c) Shared resources
- d) Paging

Answer: b

19. What is a safe state in the context of deadlock avoidance?

- a) A state where the system can allocate resources without entering deadlock
- b) A state where processes wait for resources indefinitely
- c) A state where all processes are terminated
- d) A state where all resources are allocated to one process

Answer: a

20. Which of the following methods handles deadlocks by periodically checking for their occurrence?

- a) Deadlock avoidance
- b) Deadlock detection
- c) Deadlock prevention
- d) Mutual exclusion

Answer: b

21. What is the main idea behind deadlock prevention?

- a) To allow deadlocks but recover from them later
- b) To restrict the ways processes can request resources

- c) To ignore deadlocks and hope they don't occur
- d) To preempt processes to avoid deadlocks

Answer: b

22. In the context of paging, what does a page frame refer to?

- a) A fixed-size block of memory allocated to processes
- b) A variable-size segment of memory
- c) A hardware-based solution to process synchronization
- d) A section of code that accesses shared resources

Answer: a

23. Which of the following is an advantage of paging in memory management?

- a) It increases internal fragmentation
- b) It allows dynamic memory allocation
- c) It reduces the complexity of process synchronization
- d) It ensures processes run faster

Answer: b

24. In deadlock detection, what is the resource allocation graph used for?

- a) To track process execution times
- b) To determine if a deadlock has occurred
- c) To allocate memory dynamically
- d) To swap processes between memory and disk

Answer: b

25. What is the purpose of the "turn" variable in Peterson's solution?

- a) It indicates which process should enter the critical section**
- b) It tracks the number of processes in the system**
- c) It manages memory allocation for processes**
- d) It handles the swapping of pages**

Answer: a

26. Which of the following is an essential condition for deadlock to occur?

- a) No process can be preempted**
- b) Resources are allocated dynamically**
- c) Processes can freely communicate**
- d) Processes share memory segments**

Answer: a

27. What is a deadlock?

- a) A situation where two processes exchange data**
- b) A condition where a set of processes cannot proceed because they are waiting on each other**
- c) A method to improve memory allocation**
- d) A strategy to manage process communication**

Answer: b

28. What is mutual exclusion in the context of synchronization?

- a) A condition where only one process can access a shared resource at a time**

- b) A strategy to prevent deadlock
- c) A technique for memory management
- d) A method to synchronize I/O devices

Answer: a

29. Which of the following prevents the circular wait condition in deadlock?

- a) Allowing processes to request all resources at once
- b) Using preemption to allocate resources
- c) Allocating resources in a circular manner
- d) Allowing processes to communicate directly

Answer: a

30. What is swapping in memory management?

- a) Swapping processes between CPU and I/O
- b) Moving pages between main memory and disk
- c) Allocating all resources to one process
- d) Allocating resources dynamically based on demand

Answer: b

31. What role does the "flag" variable play in Peterson's solution?

- a) It tracks the number of processes
- b) It determines which process wants to enter the critical section
- c) It is used for memory allocation
- d) It swaps pages between memory and disk

Answer: b

32. In deadlock avoidance, what is a "safe sequence"?

- a) A sequence where no process can be preempted**
- b) A sequence of resource allocation that guarantees deadlock avoidance**
- c) A scheduling order where all processes run without I/O**
- d) A condition where processes access shared resources sequentially**

Answer: b

33. What is the role of the operating system in handling deadlocks?

- a) It prevents deadlocks by blocking resource allocation**
- b) It detects, prevents, or recovers from deadlocks**
- c) It allows processes to access resources freely**
- d) It terminates processes that are running too long**

Answer: b

34. Which of the following synchronization tools can be used to avoid the critical section problem?

- a) Virtual memory**
- b) Paging**
- c) Semaphores**
- d) Swapping**

Answer: c

35. Which of the following is a disadvantage of deadlock detection and recovery?

- a) It requires that processes be terminated
- b) It increases system performance
- c) It ensures no deadlocks occur
- d) It allows resources to be allocated indefinitely

Answer: a

36. In memory management, what is the purpose of the "frame table"?

- a) To track allocated pages
- b) To manage the size of page files
- c) To allocate resources in a safe sequence
- d) To handle process synchronization

Answer: a

37. What is preemption in deadlock prevention?

- a) Preempting processes based on priority
- b) Forcing processes to release resources
- c) Swapping pages between memory and disk
- d) Allowing circular wait

Answer: b

38. What is the advantage of using a monitor for synchronization?

- a) It simplifies process communication
- b) It provides a high-level abstraction for mutual exclusion
- c) It avoids the need for paging
- d) It automatically detects deadlocks

Answer: b

39. What is the primary difference between deadlock prevention and deadlock avoidance?

- a) Deadlock prevention guarantees no deadlocks, while deadlock avoidance may allow deadlocks**
- b) Deadlock prevention uses resource allocation graphs**
- c) Deadlock avoidance allows the system to enter an unsafe state**
- d) Deadlock prevention only works for single-threaded applications**

Answer: a

40. What happens when a system is in an unsafe state?

- a) Deadlock is guaranteed to occur**
- b) The system will allocate resources indefinitely**
- c) The system may enter deadlock but it's not guaranteed**
- d) The system cannot execute any process**

Answer: c

41. What is the function of the "wait" operation in semaphores?

- a) It decreases the semaphore value by 1**
- b) It increases the semaphore value by 1**
- c) It swaps memory pages between disk and RAM**
- d) It allocates resources to processes**

Answer: a

42. What is the "signal" operation in semaphores?

- a) It decreases the semaphore value
- b) It checks for deadlocks
- c) It increases the semaphore value by 1
- d) It detects memory fragmentation

Answer: c

43. In which condition can paging cause external fragmentation?

- a) When pages are not of equal size
- b) When the page size is too small
- c) When the process size is not divisible by the page size
- d) Paging does not cause external fragmentation

Answer: d

44. What is the "hold and wait" condition in deadlock characterization?

- a) A process holding some resources can request additional resources
- b) A process holds resources indefinitely
- c) A process waits for all resources to be released
- d) A process releases resources as soon as it holds them

Answer: a

45. What is the purpose of a page table in memory management?

- a) To manage the number of processes in the system
- b) To map virtual memory addresses to physical memory addresses
- c) To detect and prevent deadlocks

d) To track the number of semaphores

Answer: b

46. How does the Banker's algorithm avoid deadlock?

- a) By allocating all resources at once**
- b) By determining a safe sequence of resource allocation**
- c) By allowing circular wait**
- d) By preempting resources from processes**

Answer: b

47. What is the role of "starvation" in deadlock management?

- a) A process never gets the resources it needs**
- b) A process holds resources indefinitely**
- c) A process cannot communicate with other processes**
- d) A process always has access to resources**

Answer: a

48. Which of the following deadlock handling methods involves killing one or more processes?

- a) Deadlock prevention**
- b) Deadlock detection**
- c) Deadlock recovery**
- d) Deadlock avoidance**

Answer: c

49. Which of the following conditions is necessary for deadlock avoidance?

- a) The system must be in a safe state**
- b) Processes must not communicate with each other**
- c) Processes must hold all resources simultaneously**
- d) Resources must be preempted frequently**

Answer: a

50. What is the main purpose of the paging mechanism?

- a) To increase system speed**
- b) To prevent deadlocks**
- c) To efficiently manage memory allocation**
- d) To synchronize process communication**

Answer: c

51. What is a semaphore?

- a) A signaling mechanism to control access to shared resources**
- b) A memory management technique**
- c) A process scheduling algorithm**
- d) A method for handling page faults**

Answer: a

52. What happens when a process performs a "wait" operation on a semaphore with a value of 0?

- a) The process is terminated**
- b) The process waits until the semaphore becomes positive**

- c) The process enters the critical section
- d) The semaphore value increases

Answer: b

53. How does deadlock detection work?

- a) By checking resource allocation graphs for cycles
- b) By preventing processes from requesting resources
- c) By preempting all resources
- d) By swapping pages between memory and disk

Answer: a

54. What is the effect of deadlock avoidance on system performance?

- a) It increases system performance by allocating resources efficiently
- b) It decreases system performance due to constant checking of safe states
- c) It has no effect on system performance
- d) It improves process communication

Answer: b

55. Which of the following memory management techniques can cause internal fragmentation?

- a) Segmentation
- b) Paging
- c) Swapping
- d) Virtual memory

Answer: b

56. What is the main characteristic of the "Dining Philosophers" problem?

- a) It illustrates how processes can cause deadlock if they don't communicate properly**
- b) It prevents memory fragmentation**
- c) It allows resource allocation dynamically**
- d) It ensures processes enter a safe state**

Answer: a

57. What happens when a system enters a deadlock?

- a) All processes continue running indefinitely**
- b) The system halts as processes wait for each other**
- c) Processes preempt resources from each other**
- d) Memory allocation is increased**

Answer: b

58. What is the "No preemption" condition in deadlock characterization?

- a) Processes cannot release resources**
- b) Resources cannot be forcibly taken from a process**
- c) Processes cannot request new resources**
- d) Processes must release resources after every operation**

Answer: b

59. Which condition is necessary for a resource allocation graph to indicate a deadlock?

- a) No available resources**

- b) A cycle in the graph
- c) A deadlock detection algorithm
- d) Multiple semaphores

Answer: b

60. What is the primary goal of memory swapping?

- a) To improve process synchronization
- b) To move processes between main memory and secondary storage
- c) To prevent deadlocks
- d) To handle resource allocation dynamically

Answer: b

61. What does the term "bounded buffer" refer to in process synchronization?

- a) A buffer that allows unlimited storage
- b) A buffer with a fixed size limit for producer-consumer processes
- c) A buffer used for paging
- d) A memory management technique

Answer: b

62. In which of the following situations is a monitor used?

- a) To manage shared memory allocation
- b) To enforce mutual exclusion in concurrent processes
- c) To preempt processes during deadlock
- d) To handle page faults in memory management

Answer: b

63. In the dining philosophers problem, what can cause deadlock?

- a) All philosophers pick up their left forks simultaneously**
- b) Only one philosopher is allowed to eat at a time**
- c) Philosophers randomly select which fork to pick up first**
- d) Philosophers wait indefinitely for both forks**

Answer: d

64. What is the role of "test and set" in synchronization hardware?

- a) It swaps memory pages between main memory and disk**
- b) It checks and sets a lock in one atomic operation**
- c) It allocates resources to processes dynamically**
- d) It ensures deadlocks do not occur**

Answer: b

65. What does the term "busy waiting" refer to?

- a) A process that constantly checks a condition until it is satisfied**
- b) A method to prevent deadlock**
- c) A technique used in paging**
- d) A memory management strategy**

Answer: a

66. Which of the following is a condition for mutual exclusion?

- a) Processes can access shared resources simultaneously
- b) Only one process can execute in the critical section at a time
- c) Processes wait indefinitely for resources
- d) Multiple processes are preempted to access shared resources

Answer: b

67. What does the "circular wait" condition describe in deadlock?

- a) Processes wait on each other in a cycle for resources
- b) Processes wait indefinitely for memory allocation
- c) A scheduling algorithm where processes are given circular access to resources
- d) A method to handle page faults

Answer: a

68. In deadlock prevention, how can the hold and wait condition be avoided?

- a) By allowing processes to hold resources indefinitely
- b) By ensuring processes request all resources at once
- c) By forcing processes to wait in a circular manner
- d) By allowing processes to request resources one at a time

Answer: b

69. What is a "wait-for graph" used for?

- a) To track memory allocation
- b) To check for deadlocks by showing which processes are waiting for resources
- c) To determine the order in which pages are swapped
- d) To allocate resources in a safe sequence

Answer: b

70. How does deadlock recovery using process termination work?

- a) Processes are allowed to run indefinitely**
- b) The system terminates all processes and restarts them**
- c) One or more processes involved in the deadlock are terminated**
- d) The system preempts resources from all processes**

Answer: c

71. In memory management, what is a page fault?

- a) When a requested page is not found in main memory**
- b) When a page is swapped between main memory and disk**
- c) When a process accesses an invalid memory address**
- d) When two processes access the same page simultaneously**

Answer: a

72. What is the purpose of using segmentation in memory management?

- a) To divide memory into equal-sized pages**
- b) To divide a program into logical segments based on function**
- c) To detect deadlocks in memory allocation**
- d) To allocate memory based on process priorities**

Answer: b

73. How does the "Banker's algorithm" ensure deadlock avoidance?

- a) By terminating all processes involved in a deadlock
- b) By preempting resources from processes
- c) By allocating resources only if the system remains in a safe state
- d) By swapping processes between main memory and disk

Answer: c

74. Which condition is violated if a resource can be forcibly taken away from a process?

- a) Mutual exclusion
- b) Hold and wait
- c) Circular wait
- d) No preemption

Answer: d

75. What does "starvation" mean in process scheduling?

- a) A process never gets scheduled for execution
- b) A process releases resources too early
- c) A process enters the critical section repeatedly
- d) A process never enters deadlock

Answer: a

76. What is the difference between internal and external fragmentation?

- a) Internal fragmentation occurs within a process; external fragmentation occurs between processes
- b) Internal fragmentation wastes space inside memory partitions; external fragmentation wastes space outside partitions

- c) Internal fragmentation occurs during deadlock; external fragmentation occurs during paging
- d) Internal fragmentation happens with virtual memory; external fragmentation happens with physical memory

Answer: b

77. What does the "turn" variable control in Peterson's solution?

- a) It controls which process can enter the critical section
- b) It manages the memory allocation for processes
- c) It tracks the number of pages in a page table
- d) It signals deadlock occurrence

Answer: a

78. In process synchronization, what is the "Readers-Writers" problem?

- a) A problem where processes must alternate between reading and writing shared data
- b) A memory management technique
- c) A synchronization issue where multiple readers or writers want to access shared data
- d) A deadlock prevention algorithm

Answer: c

79. What is the main purpose of swapping in operating systems?

- a) To allocate memory in fixed-size pages
- b) To move processes between main memory and disk when memory is full
- c) To detect deadlock conditions
- d) To synchronize processes in the critical section

Answer: b

80. What is the primary challenge in the "Dining Philosophers" problem?

- a) Avoiding memory fragmentation**
- b) Preventing deadlock by managing resource sharing**
- c) Ensuring mutual exclusion in the critical section**
- d) Allocating resources dynamically**

Answer: b